

# Module 06: SQL Joins and Set Operations

### SQL Joins Architecture & Set Operations Ecosystem

Inner, Natural, Self, Left/Right/Full Outer Joins & UNION / INTERSECT / MINUS

#### 1. SQL Join Classifications

##### INNER JOIN / EQUI JOIN

`SELECT * FROM Emp E INNER JOIN Dept D ON E.DeptID = D.DeptID;`  
Returns only rows where the join predicate condition evaluates to TRUE.

##### NATURAL JOIN & SELF JOIN

`SELECT * FROM Emp NATURAL JOIN Dept; (Auto matches same column name)`  
`SELECT E.Name, M.Name AS Manager FROM Emp E JOIN Emp M ON E.MgrID=M.EmpID;`  
Self Join joins a table with itself using distinct table aliases.

##### OUTER JOINS (Handling Dangling Records)

1. LEFT OUTER JOIN: All rows from Left table + matching Right (or NULL).  
2. RIGHT OUTER JOIN: All rows from Right table + matching Left (or NULL).  
3. FULL OUTER JOIN: All rows from both tables combined.

`SELECT E.Name, D.DeptName FROM Emp E  
LEFT OUTER JOIN Dept D ON E.DeptID = D.DeptID;`

#### 2. SQL Set Operations

##### Mandatory Rule: Union Compatibility

Both queries must project identical column counts with matching data types!

##### UNION vs UNION ALL

- UNION: Combines results and removes duplicate rows (Internal sort).
- UNION ALL: Combines results and preserves duplicates. Much faster!

`SELECT City FROM Customers UNION ALL SELECT City FROM Suppliers;`

##### INTERSECT

Returns only rows common to both query results.

`SELECT CustID FROM Active_2023 INTERSECT SELECT CustID FROM Active_2024;`

##### MINUS / EXCEPT

Returns rows in Query 1 that do NOT appear in Query 2.

`SELECT ProdID FROM All_Products MINUS SELECT ProdID FROM Sold_Products;`

Data Unification: Joins combine columns from related tables horizontally, while Set Operations append rows from union-compatible queries vertically.

## 1. SQL Outer Joins

| Join Type   | Left Table Tuples | Right Table Tuples | Unmatched Value  |
|-------------|-------------------|--------------------|------------------|
| LEFT OUTER  | 100% Retained     | Only Matches       | Padded with NULL |
| RIGHT OUTER | Only Matches      | 100% Retained      | Padded with NULL |
| FULL OUTER  | 100% Retained     | 100% Retained      | Padded with NULL |

## 2. UNION vs UNION ALL

UNION performs an internal sort-merge to eliminate duplicate rows. UNION ALL appends row streams directly without deduplication, offering superior performance.